Rocky Bayou State Park Aquatic Preserve

SEACAR Water Quality Analysis

Last compiled on 15 July, 2026

Contents

Indicators

Nutrients

Total Nitrogen - Discrete

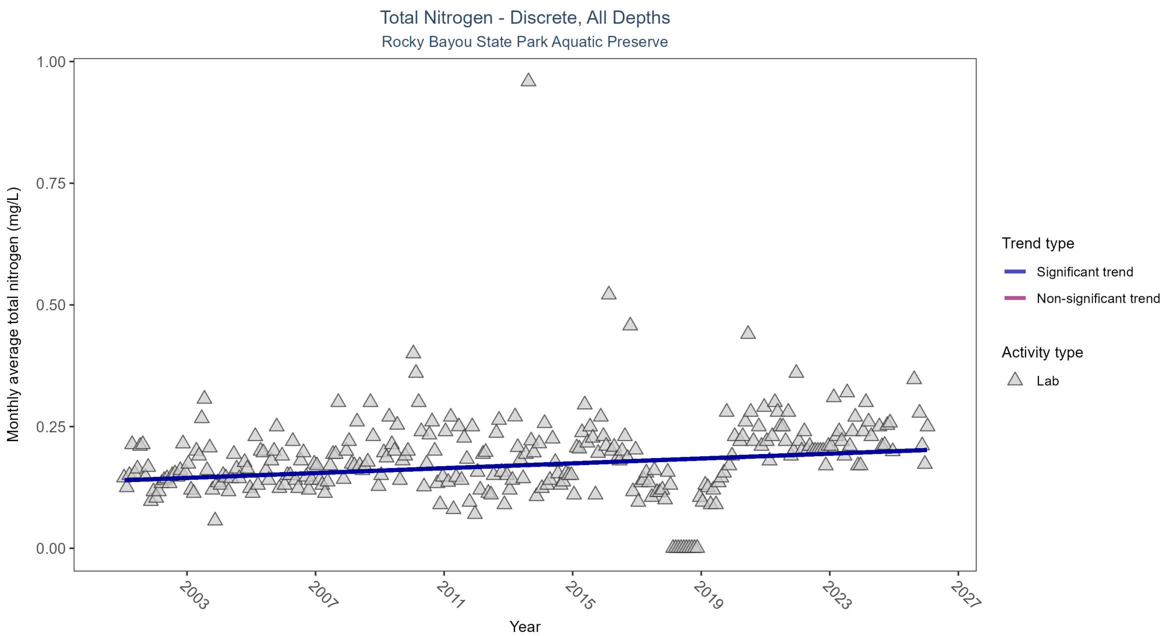

Figure 1: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 1: Monthly average total nitrogen increased by less than 0.01 mg/L per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	646	26	2001 - 2026	0.2	0.19386	0.13959	0.00249	0

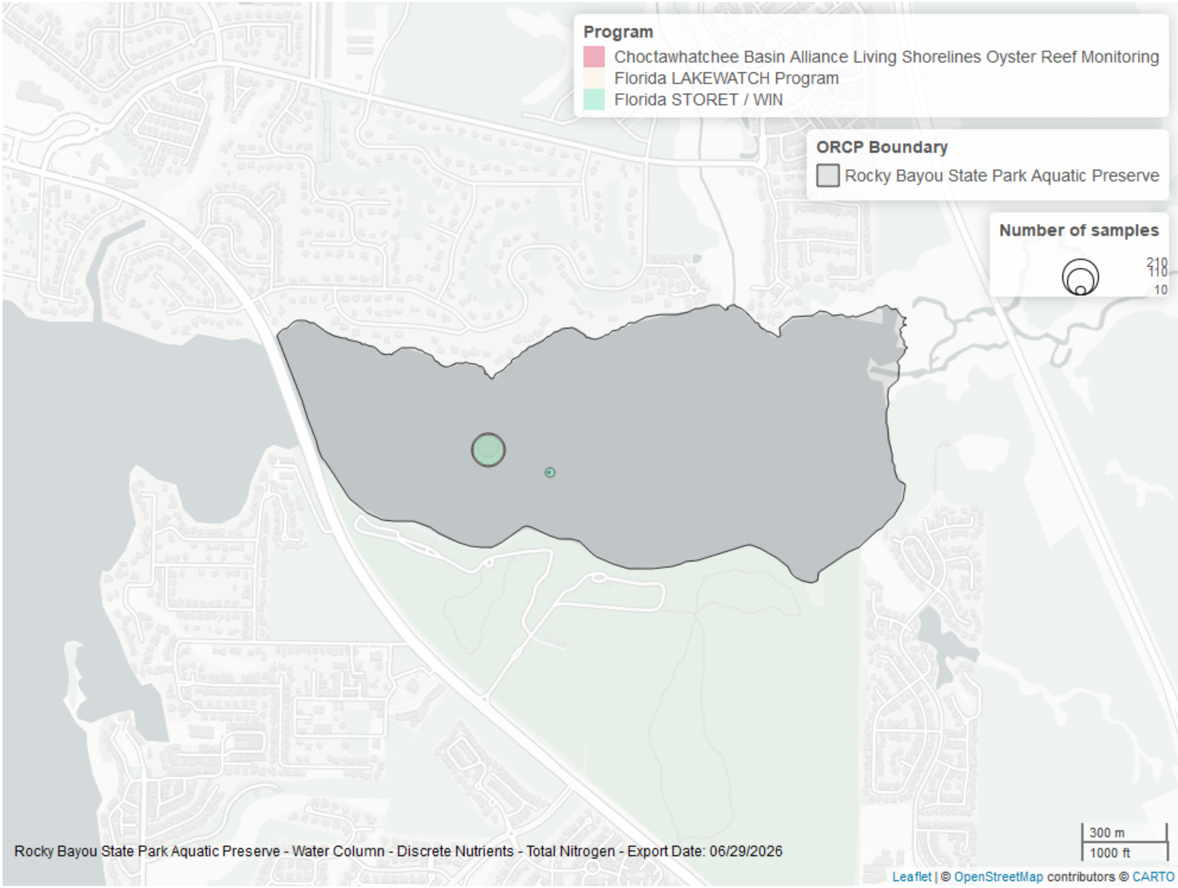

Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Rocky Bayou State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Phosphorus - Discrete

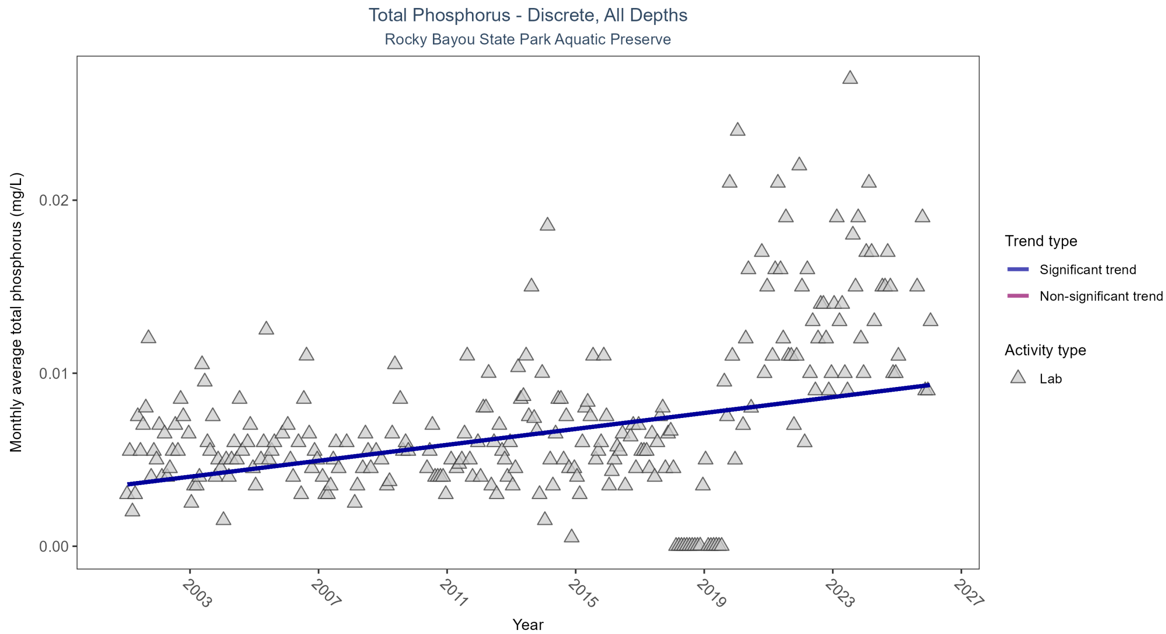

Figure 3: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 2: Monthly average total phosphorus increased by less than 0.01 mg/L per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	469	26	2001 - 2026	0.007	0.2966	0.00356	0.00023	0

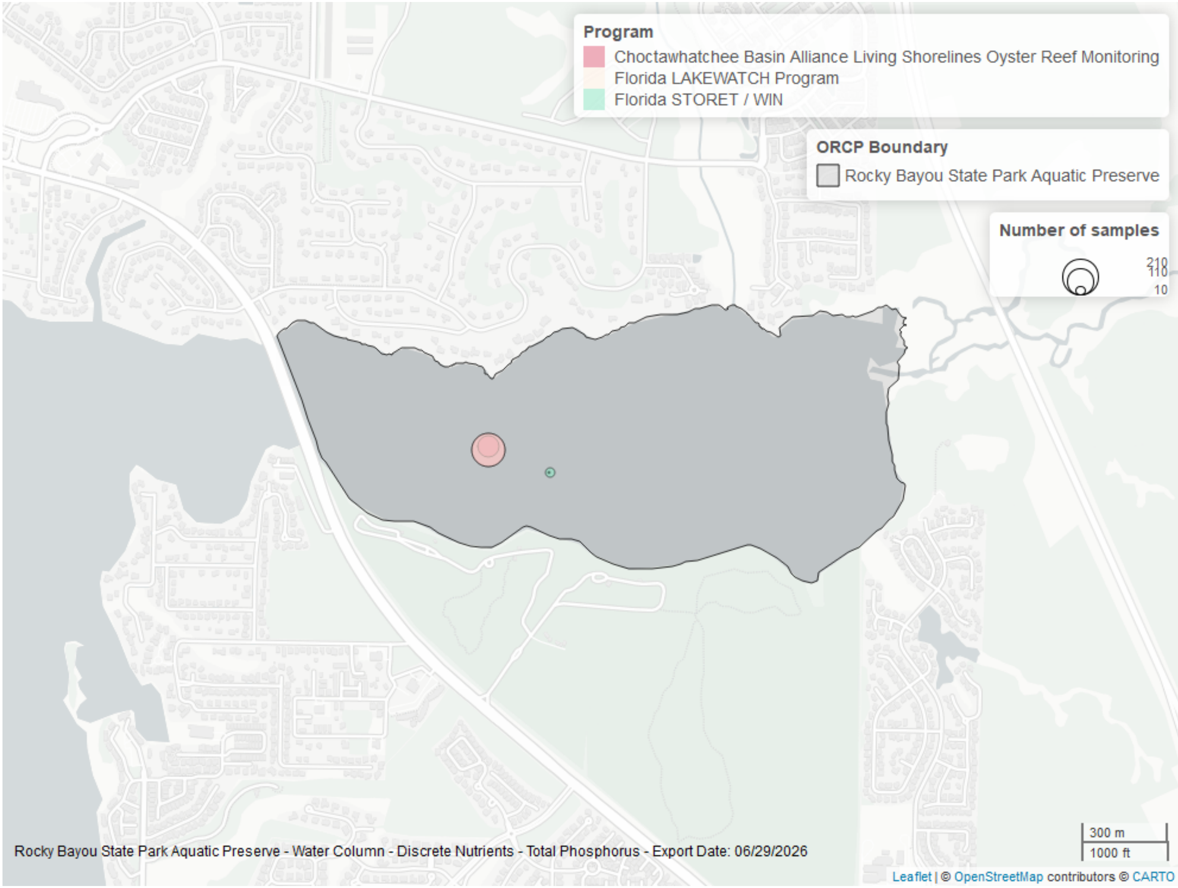

Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Rocky Bayou State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Quality

Dissolved Oxygen - Discrete

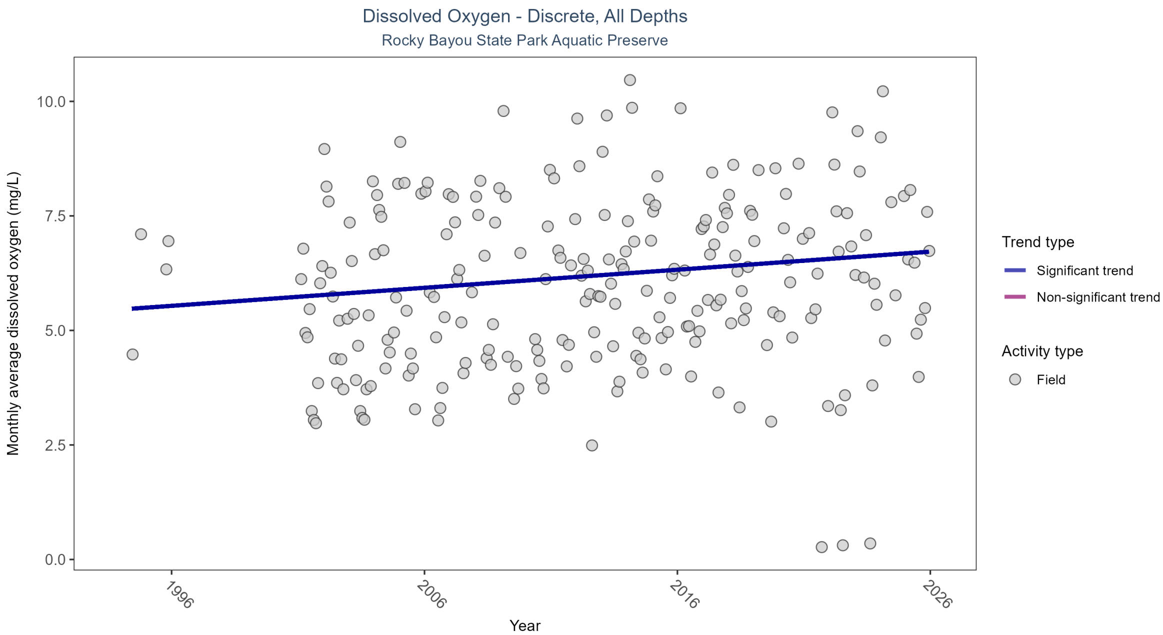

Figure 5: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 3: Monthly average dissolved oxygen increased by 0.04 mg/L per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	739	27	1994 - 2025	6.68	0.15935	5.4579	0.03942	0.0003

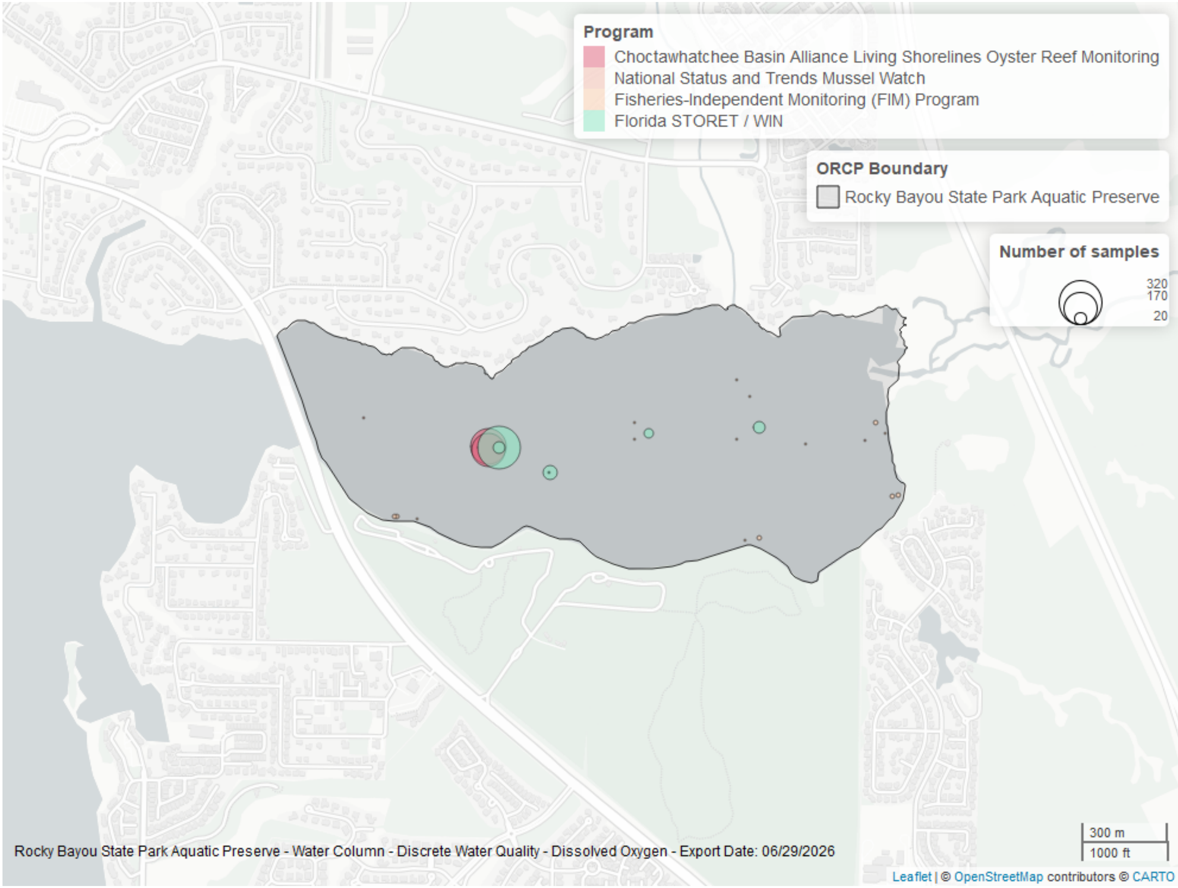

Figure 6: Map showing location of discrete water quality sampling locations within the boundaries of *Rocky Bayou State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Discrete

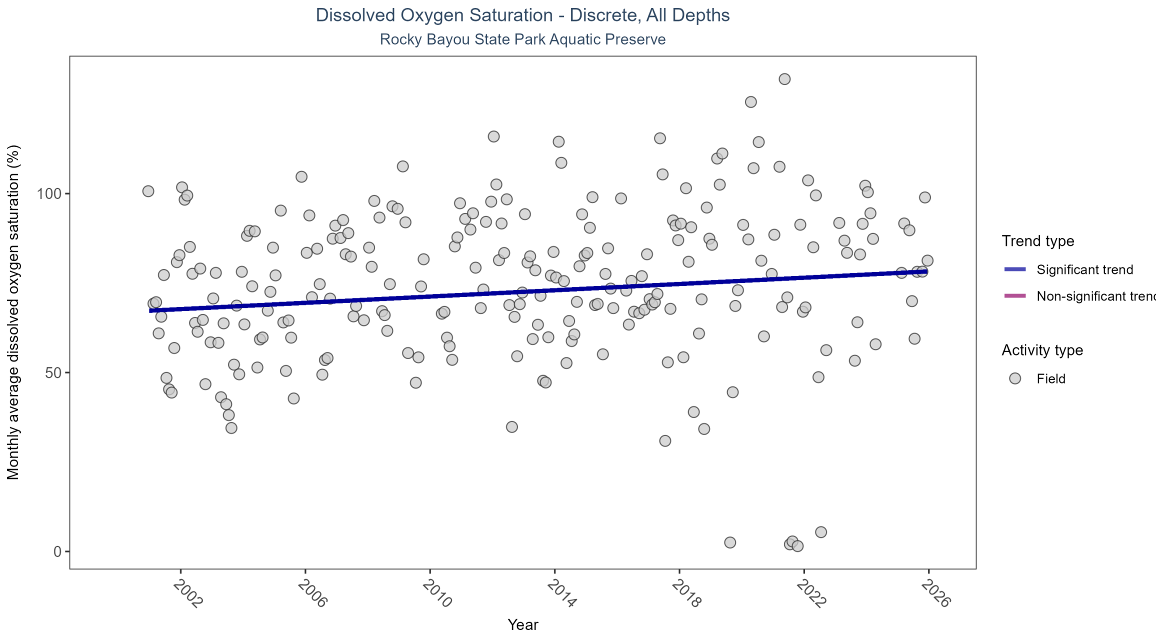

Figure 7: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

\begin{table}[H] \caption{Monthly average dissolved oxygen saturation increased by 0.44% per year.}

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	665	26	2000 - 2025	89	0.12065	66.84212	0.43893	0.009

\end{table}

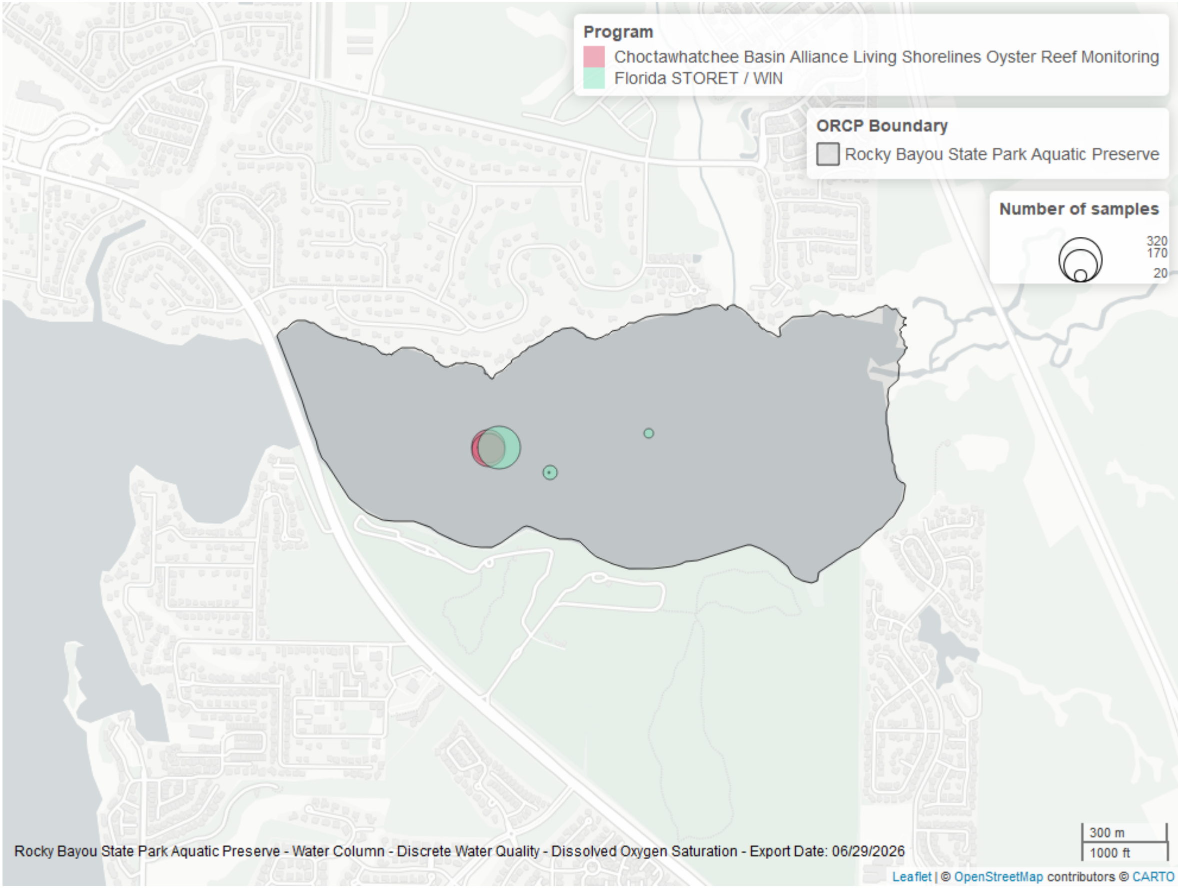

Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Rocky Bayou State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Discrete

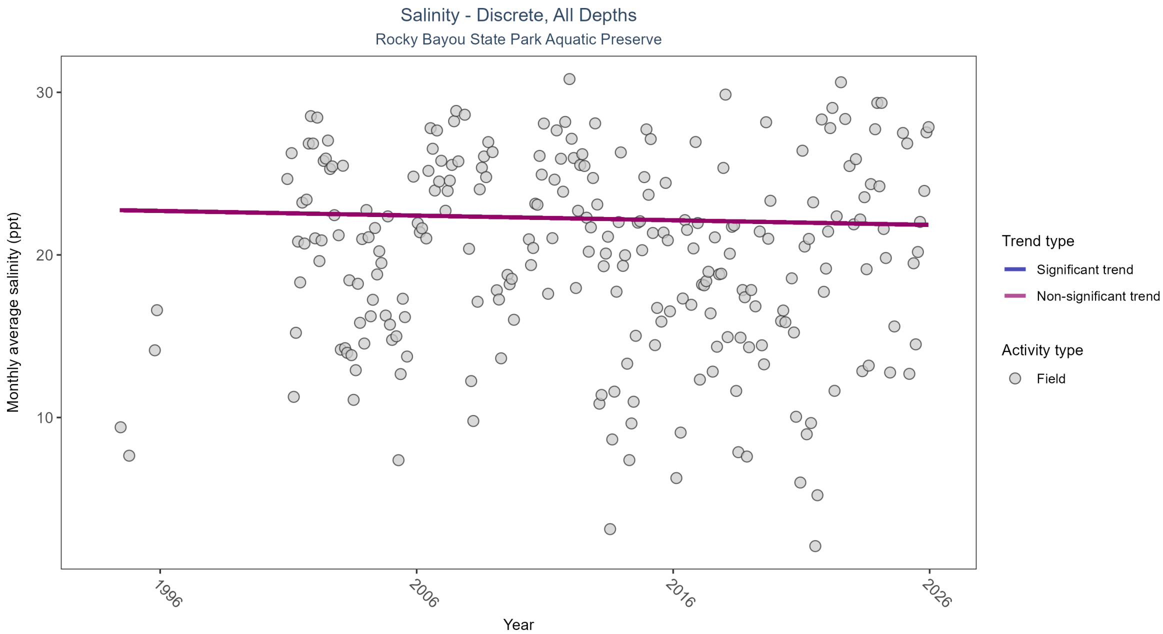

Figure 9: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 4: Salinity showed no detectable trend between 1994 and 2025.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	No detectable trend	769	28	1994 - 2025	21.63	-0.01488	22.76271	-0.02867	0.6352

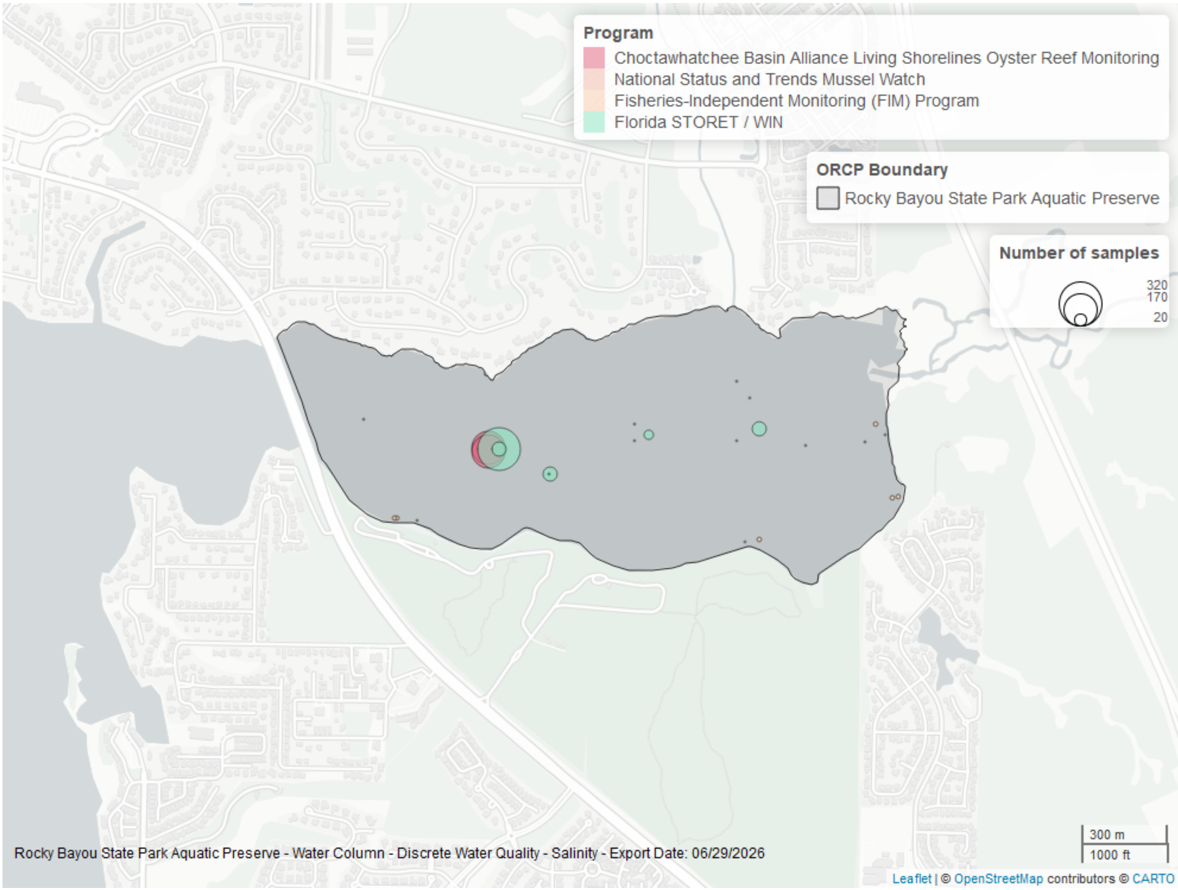

Figure 10: Map showing location of discrete water quality sampling locations within the boundaries of *Rocky Bayou State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Discrete

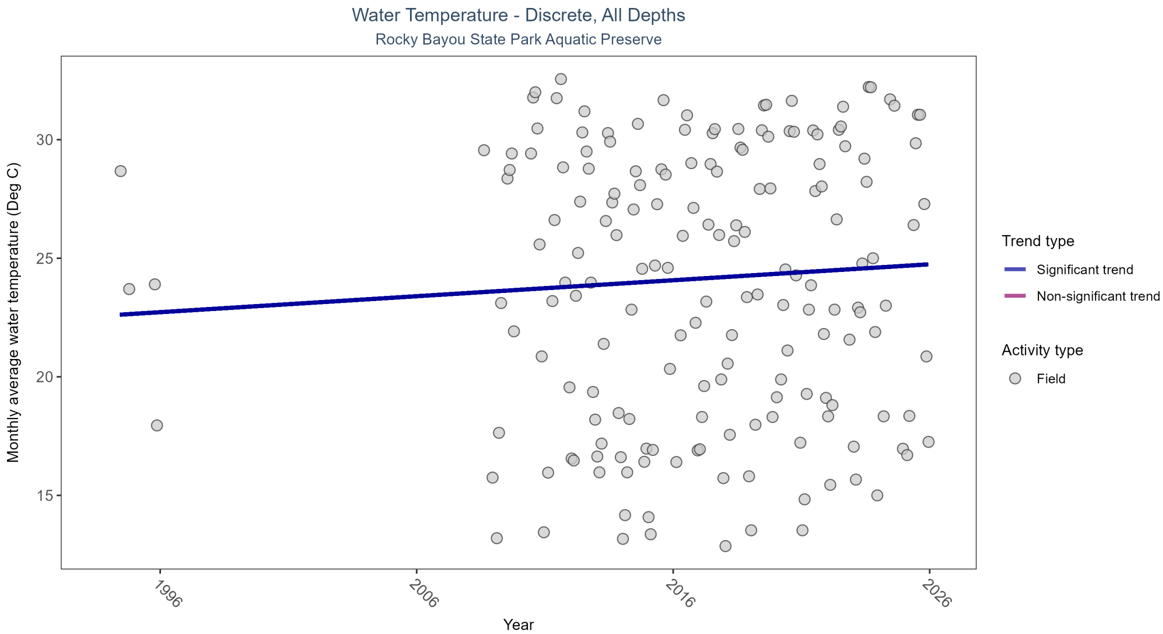

Figure 11: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 5: Monthly average water temperature increased by 0.07Deg C per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	524	20	1994 - 2025	24.83333	0.1684	22.59075	0.06733	0.0023

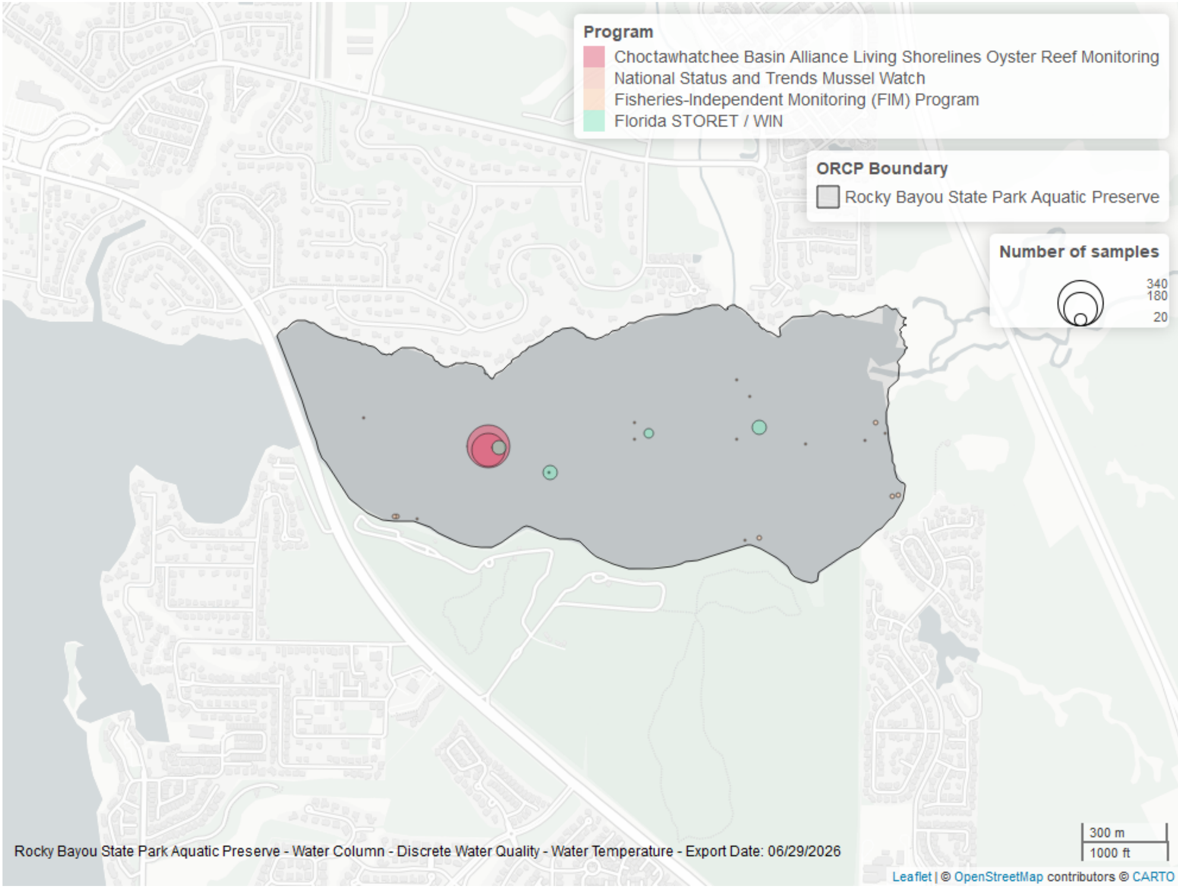

Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Rocky Bayou State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Discrete

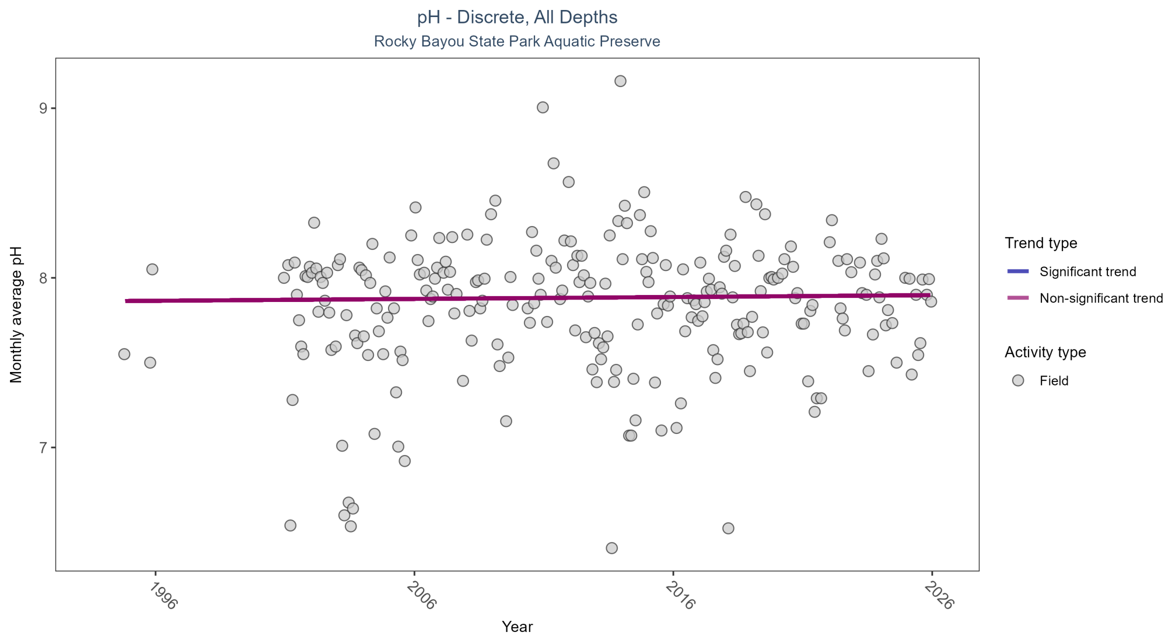

Figure 13: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 6: pH showed no detectable trend between 1994 and 2025.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	767	28	1994 - 2025	7.9	0.02089	7.86245	0.00111	0.6141

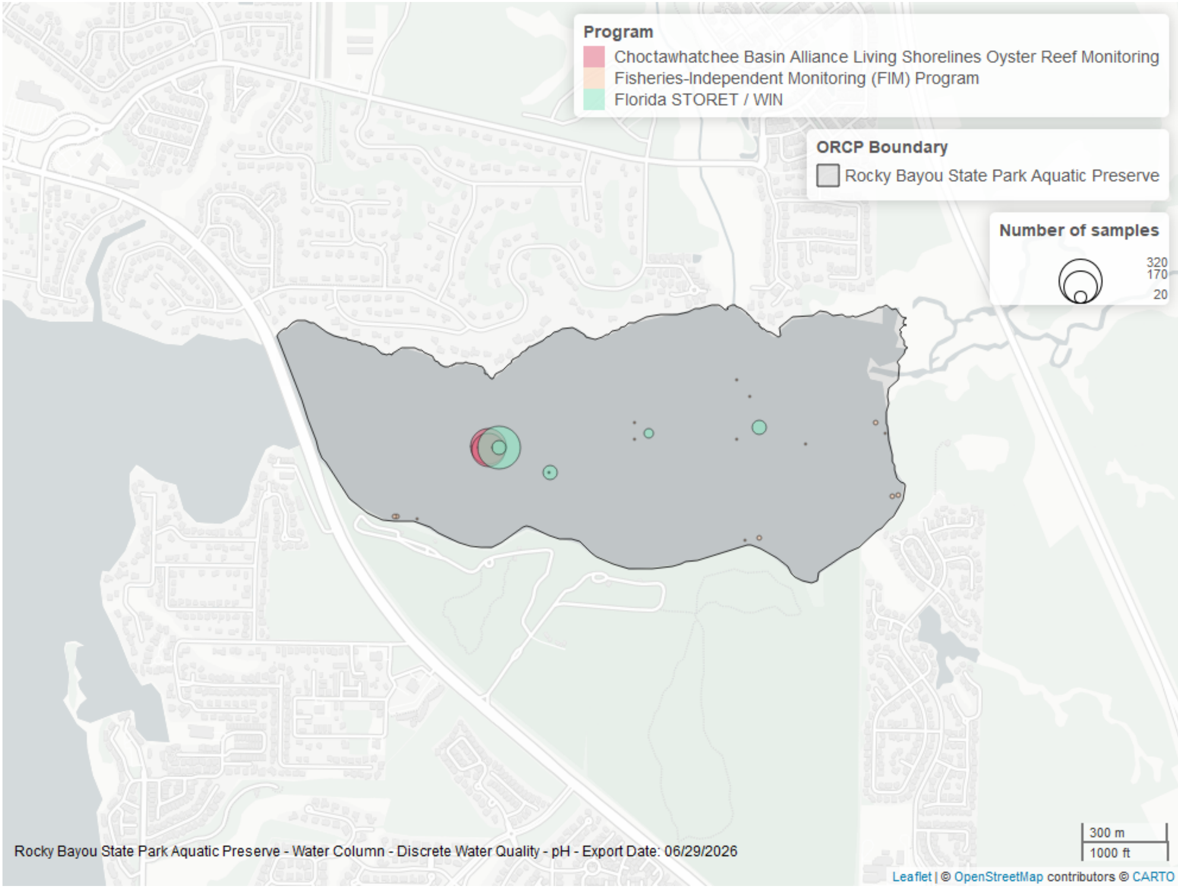

Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Rocky Bayou State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Clarity

Turbidity - Discrete

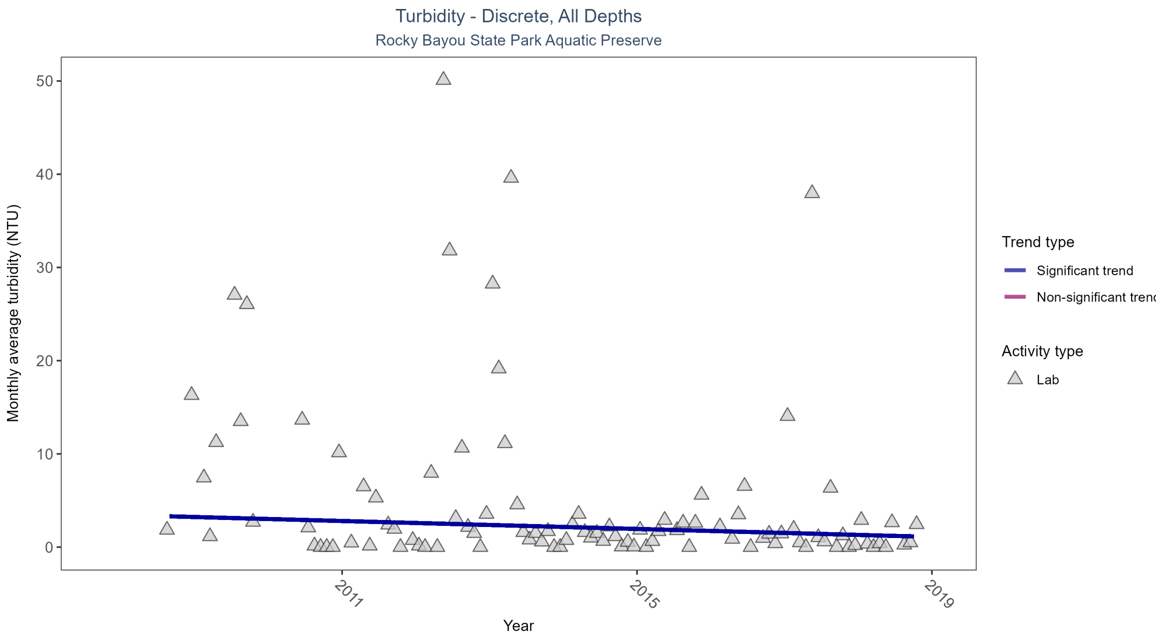

Figure 15: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 7: Monthly average turbidity decreased by 0.21 NTU per year, indicating an increase in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	222	11	2008 - 2018	1	-0.2377	3.44167	-0.21339	0.0034

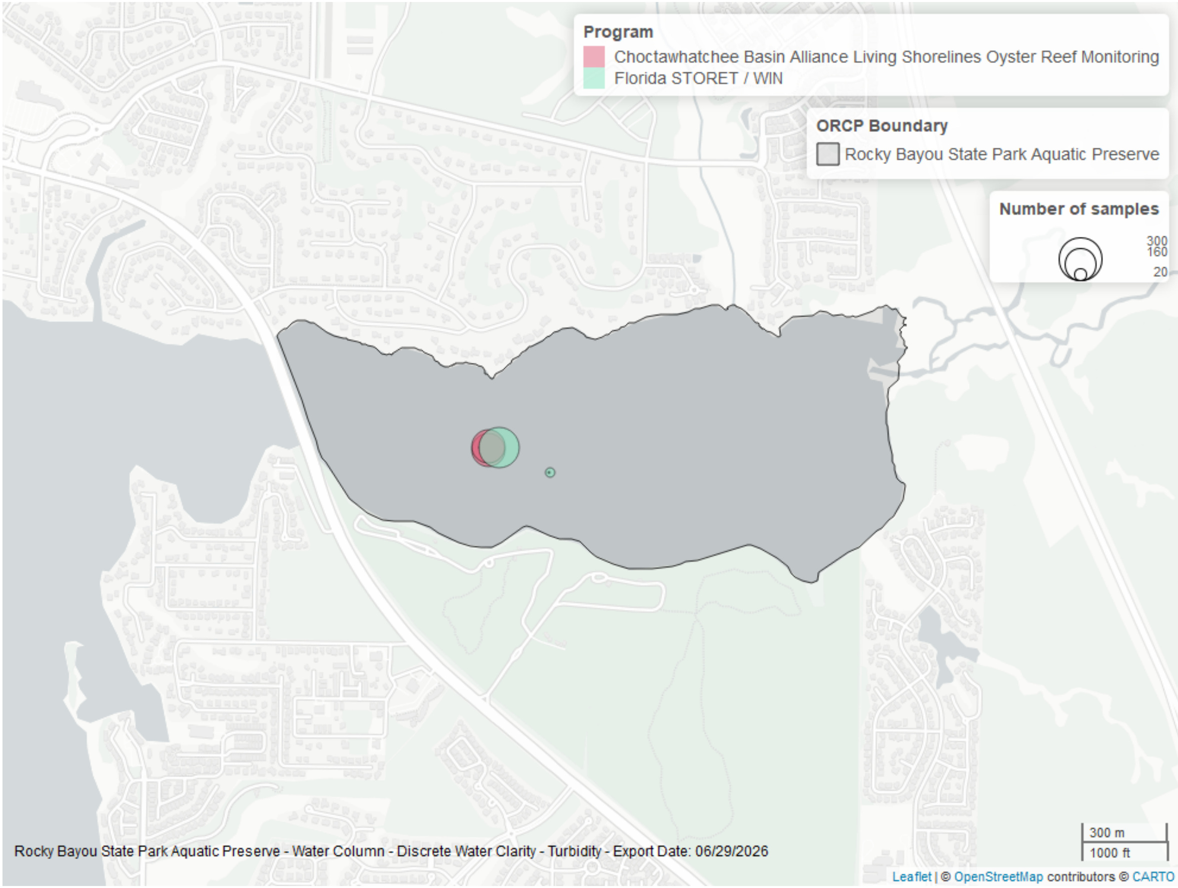

Figure 16: Map showing location of discrete water quality sampling locations within the boundaries of *Rocky Bayou State Park Aquatic Preserve.* The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Suspended Solids - Discrete

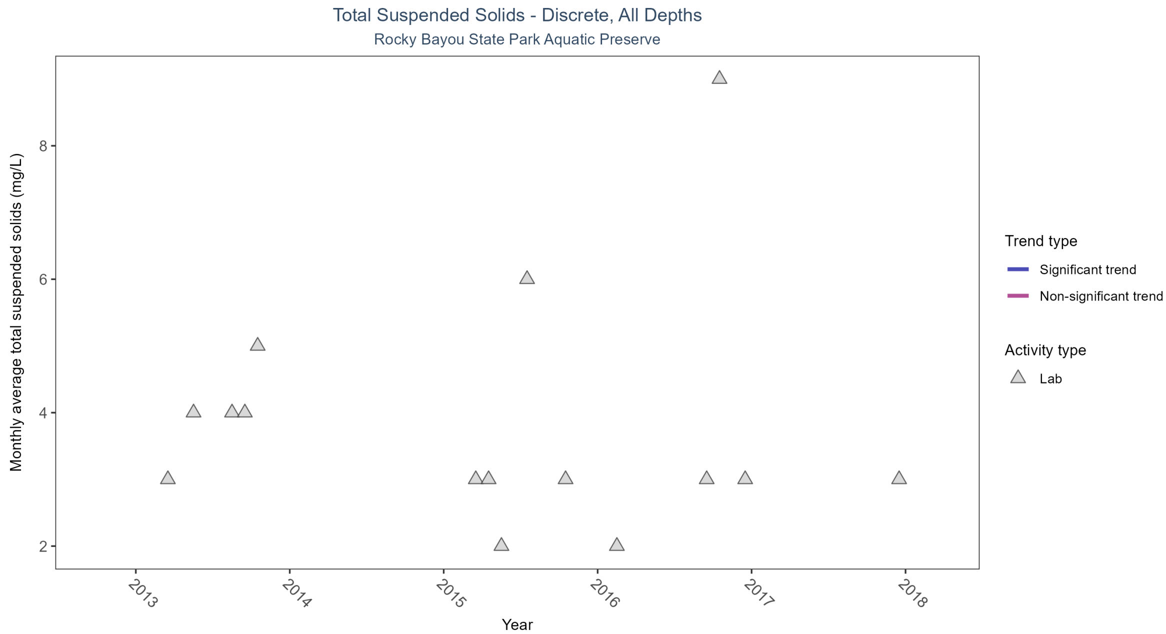

Figure 17: Scatter plot of monthly average total suspended solids (TSS) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only TSS values obtained from laboratory analyses (triangles) are included in the plot.

Table 8: There was insufficient data to fit a model for total suspended solids.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Insufficient data	17	4	2013 - 2017	3	-	-	-	-

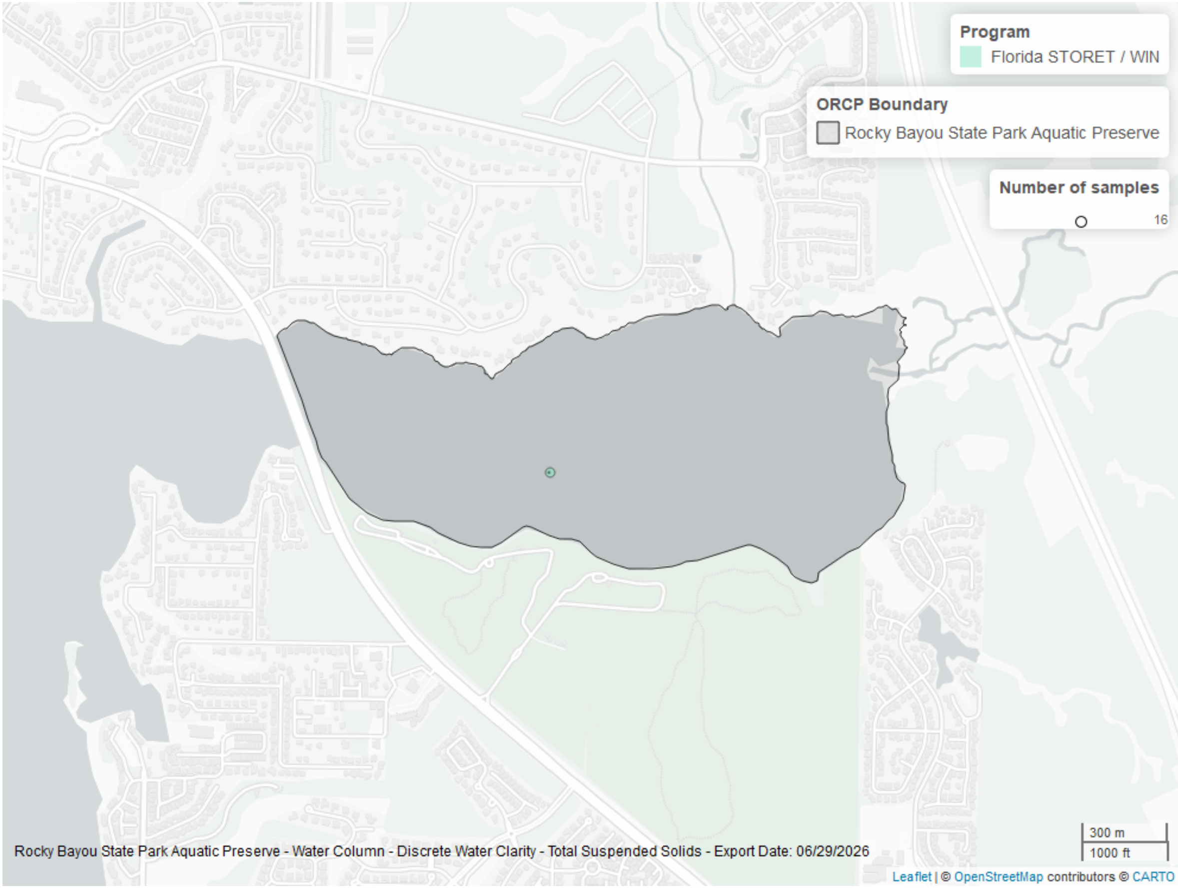

Figure 18: Map showing location of discrete water quality sampling locations within the boundaries of *Rocky Bayou State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Uncorrected for Pheophytin - Discrete

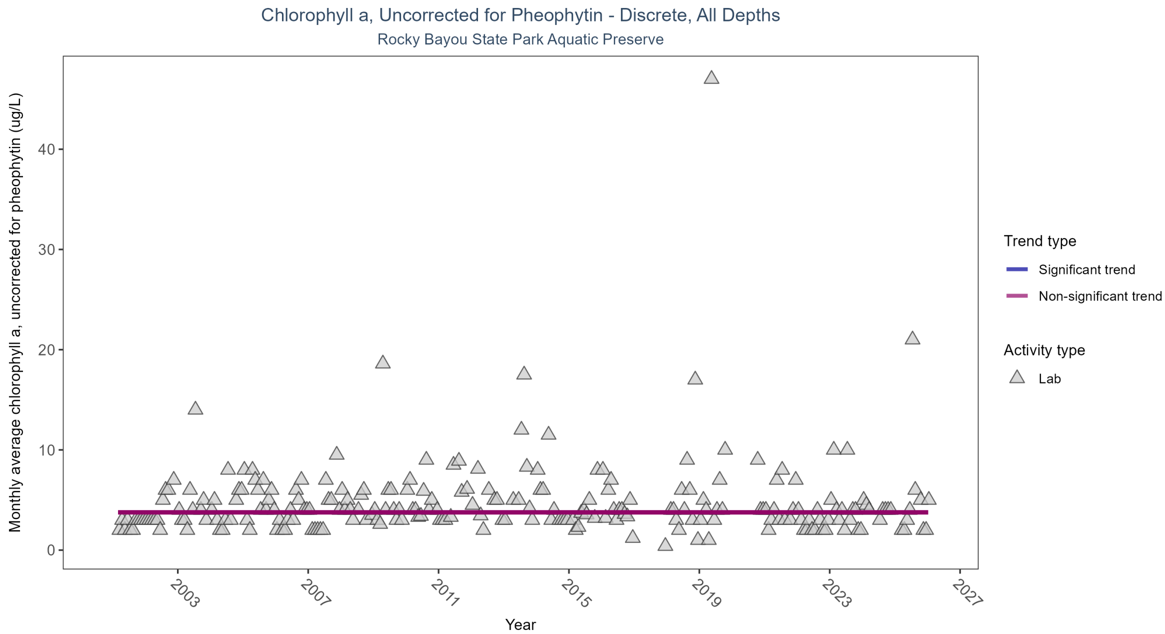

Figure 19: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 9: Chlorophyll a, uncorrected for pheophytin, showed no detectable trend between 2001 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	272	26	2001 - 2026	4	-0.00668	3.76655	0	0.9773

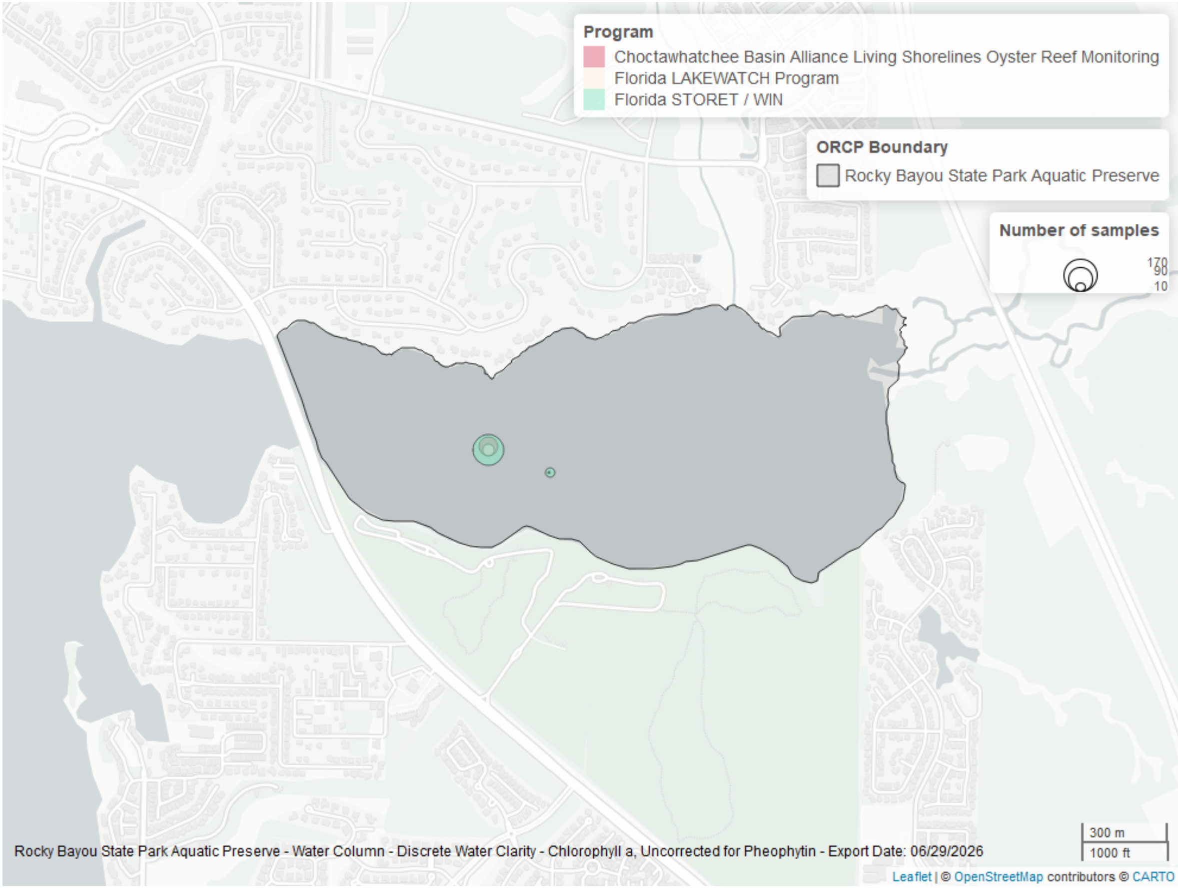

Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *Rocky Bayou State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Corrected for Pheophytin - Discrete

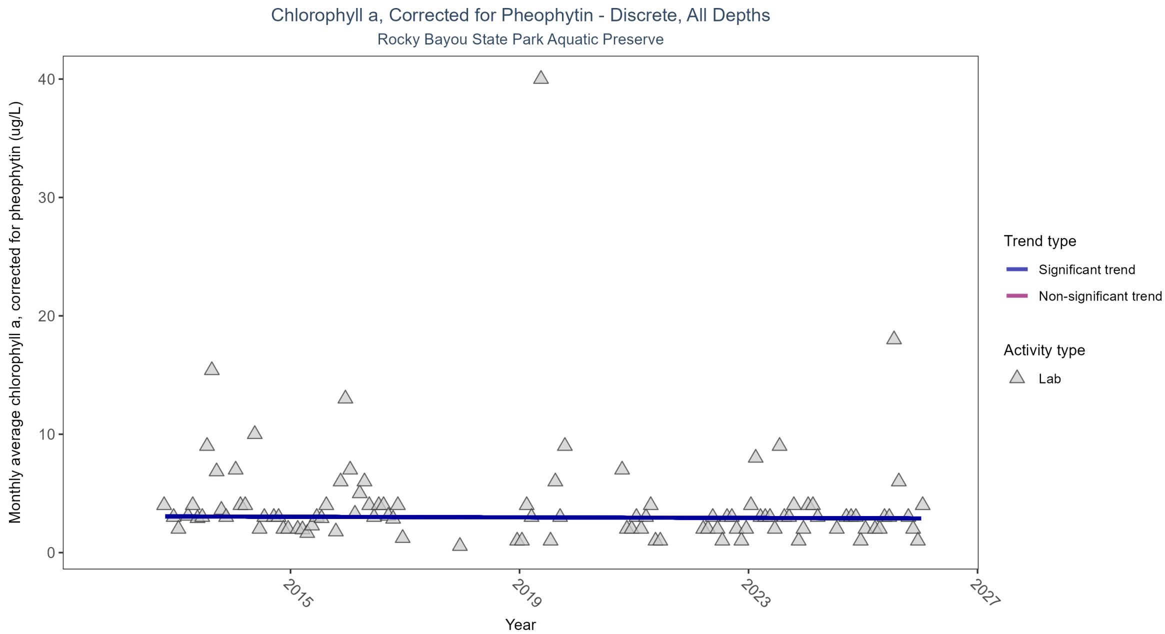

Figure 21: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 10: Monthly average chlorophyll a, corrected for pheophytin, decreased by 0.01 ugg/L per year, indicating an increase in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	120	15	2012 - 2026	3	-0.16171	3.06667	-0.0125	0.0326

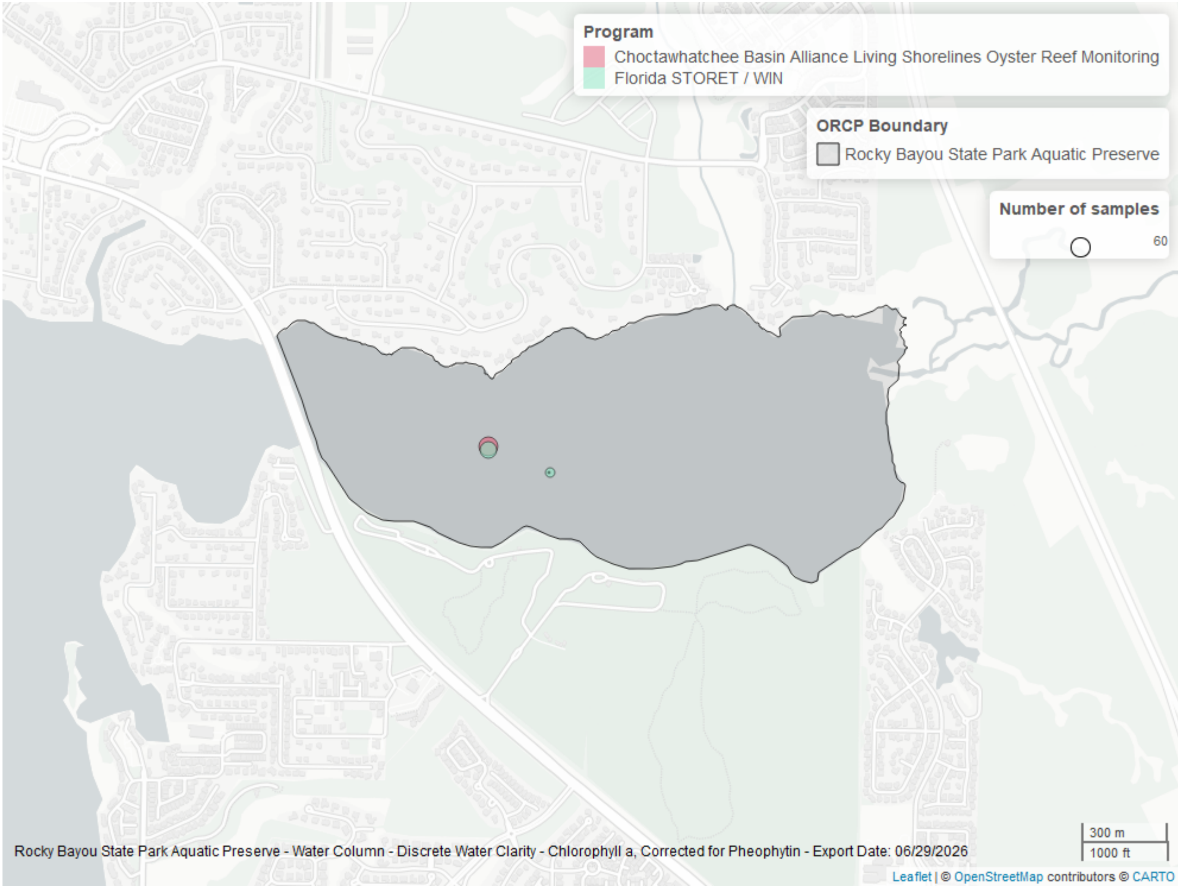

Figure 22: Map showing location of discrete water quality sampling locations within the boundaries of *Rocky Bayou State Park Aquatic Preserve.* The bubble size on the maps above reflect the amount of data available at each sampling site.

Secchi Depth - Discrete

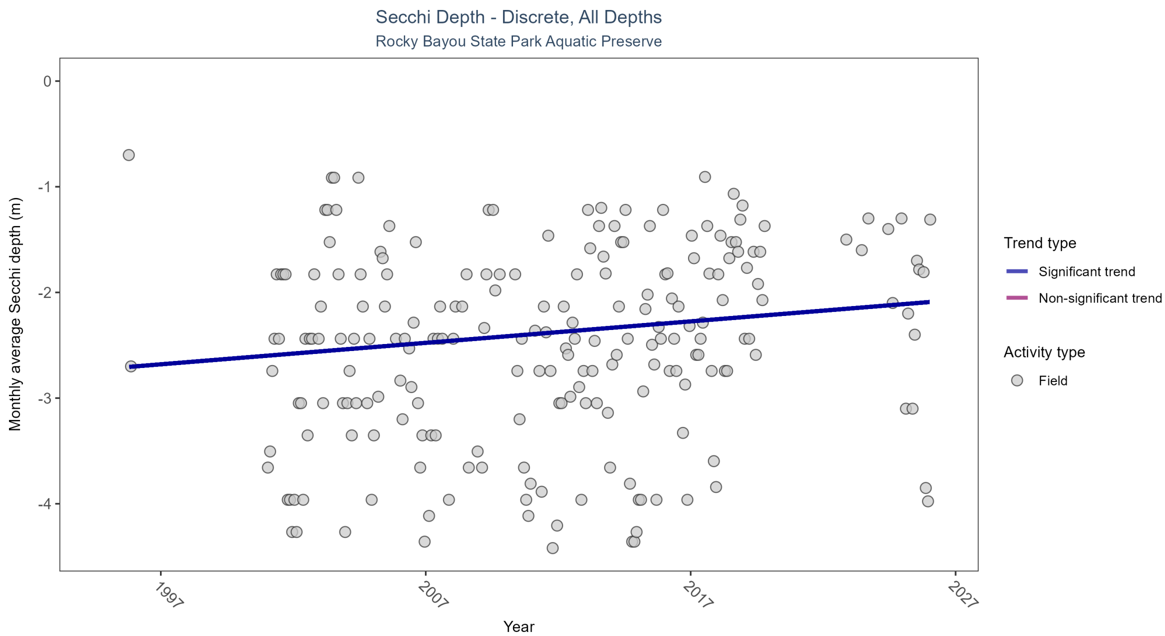

Figure 23: Scatter plot of monthly average Secchi depth over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Secchi depth is only measured in the field (circles).

Table 11: Monthly average Secchi depth became shallower by 0.02 m per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	427	25	1995 - 2026	-2.4384	0.13655	-2.72094	0.02031	0.0087

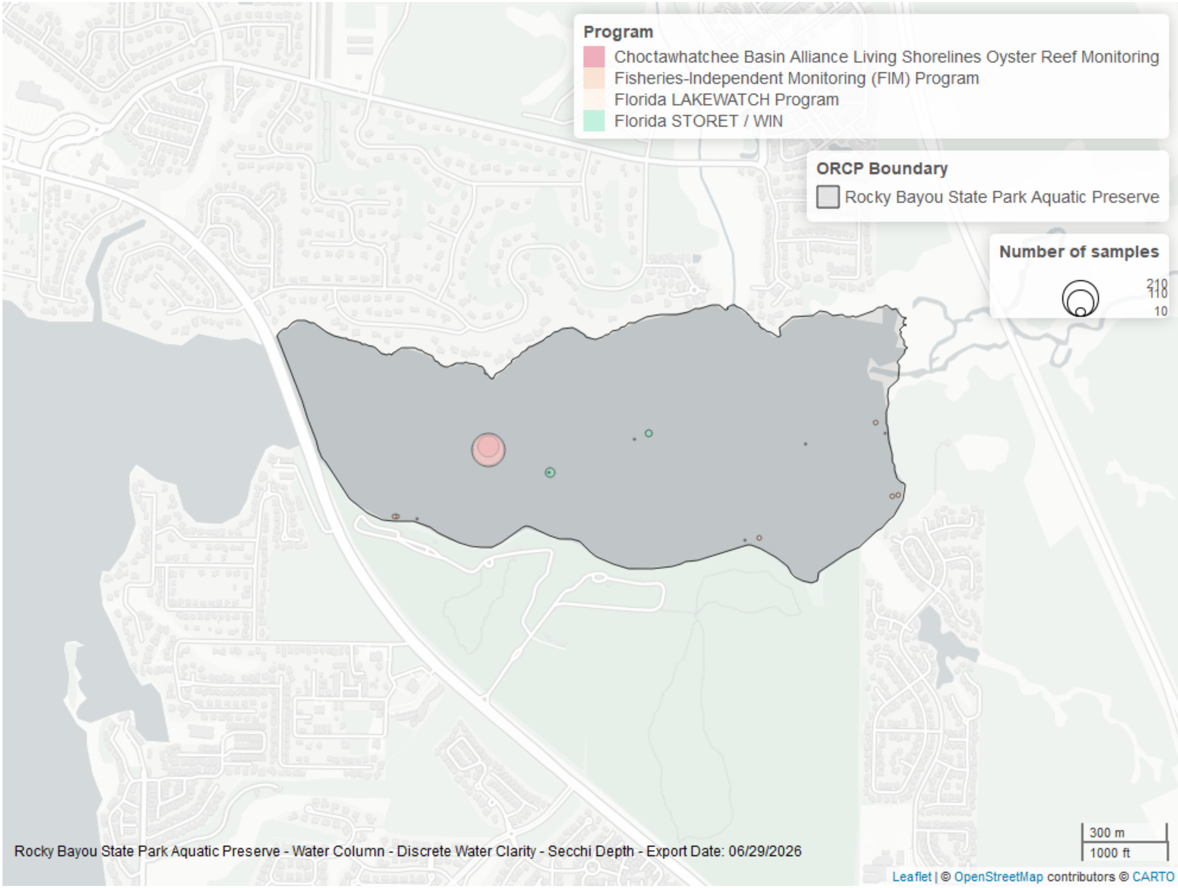

Figure 24: Map showing location of discrete water quality sampling locations within the boundaries of *Rocky Bayou State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Colored Dissolved Organic Matter - Discrete

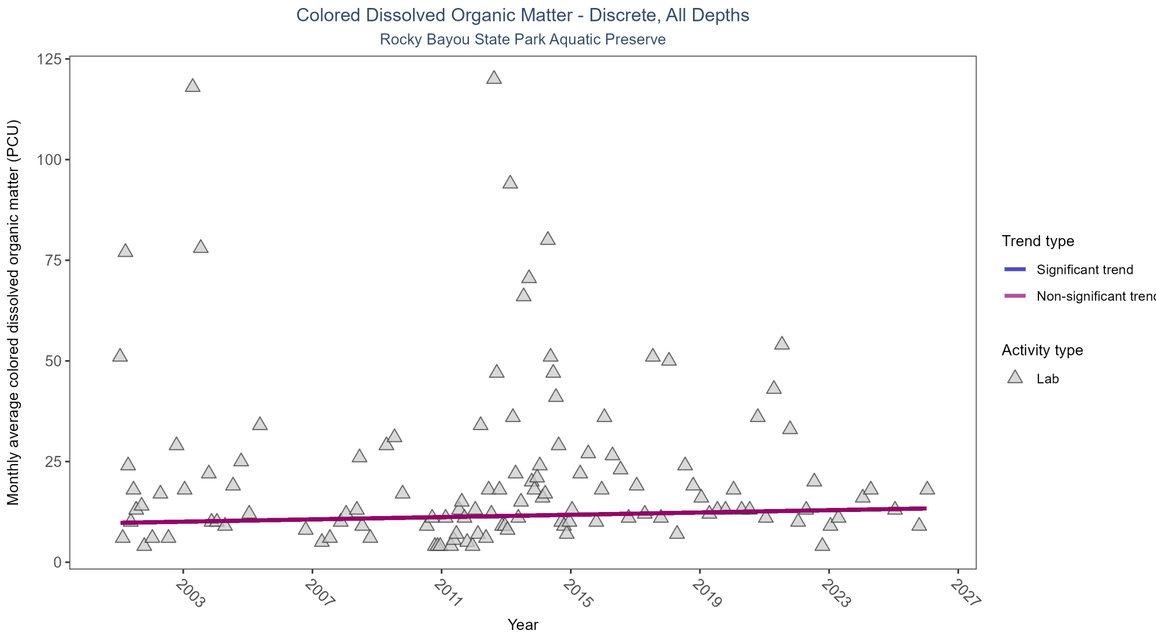

Figure 25: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 12: Colored dissolved organic matter showed no detectable trend between 2001 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	131	26	2001 - 2026	13	0.08852	9.77614	0.14286	0.3416

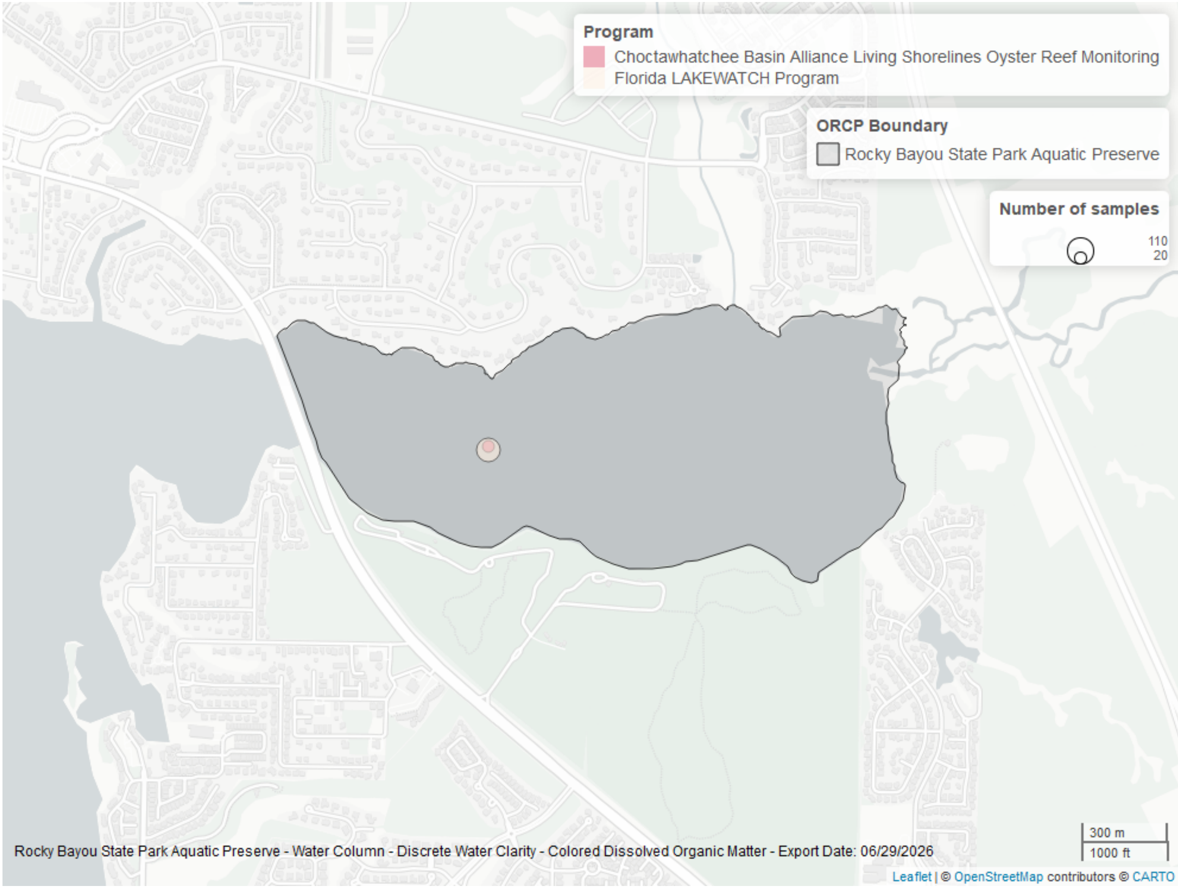

Figure 26: Map showing location of discrete water quality sampling locations within the boundaries of *Rocky Bayou State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.